Exact Finite Termination and Cell Geometry for One-Dimensional Bounded Confidence

Route-A source-local certificate HCS-C312

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Abstract

For the homogeneous one-dimensional Hegselmann–Krause map we prove order and permanent-gap decomposition, exact finite termination within $4n^3 + 2n + 2$ updates, and the complete fixed-cluster classification.

1 Order, decomposition, and finite-time theorem

Let $\varepsilon > 0$ and, after sorting the labels, set

$$N_i(t) = \{j : |x_j(t) - x_i(t)| \le \varepsilon\}, \qquad x_i(t+1) = \frac{1}{|N_i(t)|} \sum_{j \in N_i(t)} x_j(t). \tag{1}$$

The inequality is closed: a pair at distance exactly ε interacts.

Theorem 1 (Complete one-dimensional stopping atlas). *For every $n \ge 1$ and every real initial state:*

1. *order, coincident blocks, and the convex hull are preserved;*
2. *a consecutive gap greater than ε never closes and splits the dynamics into independent subsystems;*
3. *the state becomes exactly fixed by time $4n^3 + 2n + 2$;*
4. *a state is fixed iff its distinct occupied positions are separated by gaps strictly greater than ε.*

The key quantitative fact is included for clarity.

Lemma 1 (Two-step progress). *If the system is not fixed at time t, let ℓ be its leftmost nonfrozen position block. By time $t + 2$, that block has gained multiplicity, has frozen, or its position has moved right by at least $\varepsilon/(2n^2)$.*

Proof. Scale to $\varepsilon = 1$ and let r be the first strictly right neighbor of ℓ. If their neighborhoods coincide, their next averages coincide and the left block gains weight. Otherwise r has a right neighbor unseen by ℓ, hence beyond $x_\ell + 1$. Direct averaging over at most n points gives $x_r(t+1) \ge x_\ell(t) + 1/n$. If ℓ itself moved by $1/(2n)$ we are done. Otherwise their new separation is at least $1/(2n)$. If it now exceeds one, ℓ freezes; if not, the next average moves its block by at least another factor $1/n$. Rescaling proves the claim. □

Proof of Theorem 1. The average of two ordered interval neighborhoods is ordered; identical positions have identical neighborhoods. Every update lies in its neighborhood convex hull. If a consecutive gap exceeds ε, the two sides never average across it and their separate convex hulls cannot approach, proving items 1–2.

Decompose the initial state at all such gaps. A component of m agents has width at most $(m-1)\varepsilon$. Apply Lemma 1 at even times. The leftmost active position is nondecreasing; multiplicity/freeze events exhaust at most m labels, while displacement events occur at most $2m^3$. Thus the component stops within $4m^3 + 2m + 2$ updates. Independent components evolve in parallel, so the full stopping time is their maximum; since every $m_j \le n$, this is at most $4n^3 + 2n + 2$, proving item 3.

Clusters separated by strict gaps are plainly fixed. Conversely, the leftmost occupied position having any distinct neighbor within ε would average strictly to the right, contradicting fixedness. This proves item 4. □

Source lineage

References

[1] A. Bhattacharyya, M. Braverman, B. Chazelle, and H. L. Nguyen, "On the convergence of the Hegselmann–Krause system," *ITCS 2013*, 61–66. arXiv:1211.1909.